EE 533 Information Theory

Week 04

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Strings, Cartesian Products, and the Length Function

For any countable (i.e., countably infinite or finite) set $\mathcal{Y}$,

- $\mathcal{Y}_1^k$ is the set of all length k sequences (strings) of elements of $\mathcal{Y}$:

$$\mathcal{Y}_1^k := \{y_1 \dots y_k : y_j \in \mathcal{Y} \text{ for all } j \in \{1, \dots, k\}\}.$$

- $\mathcal{Y}^*$ is the set of all finite strings of elements $\mathcal{Y}$:

$$\mathcal{Y}^* := \bigcup_{j \in \mathbb{Z}_+} \mathcal{Y}_1^j.$$

  - $\mathcal{Y}^* \neq \mathcal{Y}_1^\infty$.
  - If $\mathcal{Y}$ is finite, then $\mathcal{Y}_1^k$ is finite and $\mathcal{Y}^*$ is countably infinite.
  - If $\mathcal{Y}$ is countably infinite, then both $\mathcal{Y}_1^k$ and $\mathcal{Y}^*$ are countably infinite.

The length function $\ell : \mathcal{Y}^* \to \mathbb{Z}_+$ is

$$\ell(y_1^k) = k \qquad \forall y_1^k \in \mathcal{Y}^*$$

Source Coding (Data Compression)

Definition (Source Code)

A source code for a discrete random variable X is a function of the form

$$\mathcal{C} : \mathcal{X} \to \mathcal{D}^*$$

where $\mathcal{X}$, i.e. the range of X, is a countable set and $\mathcal{D}$ is a finite set.

$\mathcal{C}(x)$ is called the codeword corresponding to x for all $x \in \mathcal{X}$.

$\ell(\mathcal{C}(x))$ is the length of $\mathcal{C}(x)$ for all $x \in \mathcal{X}$.

Definition

The expected length $L(\mathcal{C})$ of a source code $\mathcal{C}$ is defined as

$$\begin{aligned} L(\mathcal{C}) &:= \mathbf{E}\left[\ell(\mathcal{C}(\mathsf{X}))\right] \\ &= \sum\nolimits_x p_{\mathsf{X}}(x)\, \ell(\mathcal{C}(x)) \end{aligned}$$

Whenever the code $\mathcal{C}$ is clear from the context we will use $\ell(x)$ instead of $\ell(\mathcal{C}(x))$ and L instead of $L(\mathcal{C})$.

Examples

$$\mathcal{X} = \{a, b, c, d\}, \qquad p_X(x) = \begin{cases} 1/2 & x = a \\ 1/4 & x = b \\ 1/8 & x = c \\ 1/8 & x = d \end{cases} \qquad \text{and} \qquad \mathcal{C}(x) = \begin{cases} 0 & x = a \\ 10 & x = b \\ 110 & x = c \\ 111 & x = d \end{cases}$$

$$\begin{aligned} L(\mathcal{C}) &= (1/2) \times 1 + (1/4) \times 2 + (1/8) \times 3 + (1/8) \times 3 \\ &= \tfrac{7}{4} \\ H(X) &= \tfrac{7}{4} \log 2 \end{aligned}$$

$$\mathcal{X} = \mathbb{Z}_+, \qquad p_X(x) = 2^{-x} \qquad \text{and} \qquad \mathcal{C}(x) = \underbrace{11 \cdots 11}_{(x-1)\ 1's}0$$

$$\begin{aligned} L(\mathcal{C}) &= \sum_{x=1}^{\infty} (2^{-x})x \\ &= 2 \\ H(X) &= 2 \log 2 \end{aligned}$$

$$= \sum_{x=1}^{\infty} \sum_{j=1}^{x} 2^{-x} = \sum_{j=1}^{\infty} \sum_{x=j}^{\infty} 2^{-x}$$

What is wrong with $\mathcal{C}$ s.t. $\mathcal{C}(x) = 0$ for all $x \in \mathcal{X}$?

Definition

A code $\mathcal{C}$ is nonsingular iff

$$x \neq \widetilde{x} \Rightarrow \mathcal{C}(x) \neq \mathcal{C}(\widetilde{x})$$

Definition

The extension of a code $\mathcal{C} : \mathcal{X} \to \mathcal{D}^*$ is a function $\mathcal{C}^* : \mathcal{X}^* \to \mathcal{D}^*$ s.t.

$$\mathcal{C}^*(x_1 \dots x_k) = \mathcal{C}(x_1) \dots \mathcal{C}(x_k) \qquad \forall k \in \mathbb{Z}_+, x_1 \dots x_k \in \mathcal{X}_1^k$$

Definition

A code $\mathcal{C} : \mathcal{X} \to \mathcal{D}^*$ is *uniquely decodable* iff its extension $\mathcal{C}^*$ is nonsingular.

Definition

A string d_1^k is a prefix of another string $\tilde{d}_1^n$ iff $d_1^k = \tilde{d}_1^k$.

Definition

A code $\mathcal{C} : \mathcal{X} \to \mathcal{D}^*$ is a *prefix (free) code* or *instantaneous code* iff no codeword is a prefix of another codeword.

Instantaneous code $\Rightarrow$ Uniquely decodable $\Rightarrow$ Nonsingular

Instantaneous code $\Rightarrow$ Uniquely decodable $\Rightarrow$ Nonsingular

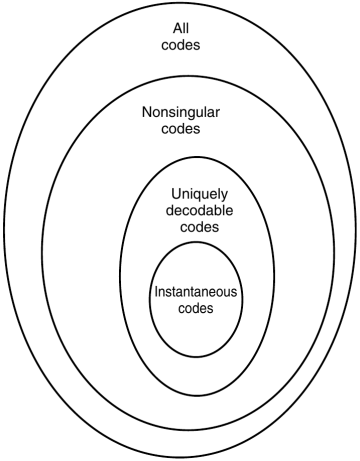


Figure: Classes of Codes [CT06, Table 5.1]

X	Singular	Nonsingular, But Not Uniquely Decodable	Uniquely Decodable, But Not Instantaneous	Instantaneous
1	0	0	10	0
2	0	010	00	10
3	0	01	11	110
4	0	10	110	111

Figure: Classes of Codes [CT06, Table 5.1]

- In the second code 010 can be 2, 14, or 31
- In the third code:
  - If the string starts with 10 the first source symbol is 1.
  - If the string starts with 00 the first source symbol is 2.
  - If the string starts with 11 followed by an even number of 0's the first source symbol is 3.
  - If the string starts with 11 followed by an even number of 0's the first source symbol is 4.

Theorem (Kraft Inequality)

For any finite set of codewords that form an instantaneous (prefix) code over an alphabet of size D, codeword lengths must satisfy the inequality

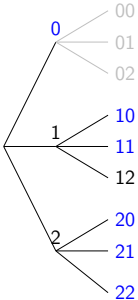
$$\sum_{j=1}^{m} D^{-l_j} \leq 1. \qquad (*)$$

Conversely given a $l_1, l_2, \ldots, l_m$ satisfying this inequality, there exists an instantaneous code with these word lengths.

Proof of (*)

- $l_{\max} = \max_i l_i$
- Label the leaves and the inner nodes of a D-ary tree of depth $l_{\max}$,
- Each codeword of length l_i corresponds to $D^{l_{\max}-l_i}$ leaves.
- Since code is instantaneous, each leaf is claimed by at most one codeword.
- $\sum_{j=1}^{m} D^{l_{\max}-l_j} \leq D^{l_{\max}} \Rightarrow (*)$

$D=3,\ l_{\max}=2$



Existence of a prefix code satisfying (*)

(a) Order l_j's non-decreasingly.

(b) Choose the smallest l_j and assign it to the smallest node of depth l_j that has not been assigned to another l_i or that has not been removed by another l_i.

(c) Remove all the descendant of the newly assigned node and repeat part (b).

Theorem (Extended Kraft Inequality)

For any countable set of codewords that form an instantaneous (prefix) code over an alphabet of size D, codeword lengths must satisfy the inequality

$$\sum_{i\in\mathbb{Z}_+} D^{-l_i} \leq 1. \qquad (**)$$

Conversely given a $l_1, l_2, \ldots$ satisfying this inequality, there exists an instantaneous code with these word lengths.

Proof. Let $g : \mathcal{D}^* \to \left\{ \left[\frac{s}{D^n}, \frac{s+1}{D^n}\right) : n \in \mathbb{Z}_+, s \in \{0, 1, \ldots, D^n - 1\}\right\}$ be

$$g(y_1^n) := \left[0.y_1 \ldots y_n, 0.y_1 \ldots y_n + D^{-n}\right) \qquad \forall y_1^n \in \mathcal{D}^*$$

- g is invertible and $y_1^n \in \mathcal{D}^*$ is a prefix of $z_1^k \in \mathcal{D}^*$ iff $g(z_1^k) \subset g(y_1^n)$.
- $\mathcal{C} \subset \mathcal{D}^*$ is an instantenous code iff elements of $g(\mathcal{C}) \subset g(\mathcal{D}^*)$ are non-overlapping. $\Rightarrow$ (**)
- If $\ell_1, \ell_2, \ldots$ is a non-decreasing sequence of positive integers satisfying (**), then $\left[0, D^{-\ell_1}\right) \in g(\mathcal{D}^*)$ and $\left(\sum_{k=1}^{t-1} D^{-\ell_k}\right) D^{\ell_t} \in \mathbb{Z}_+$ for all $t > 1$. Thus $\left[\sum_{k=1}^{t-1} D^{-\ell_k}, \sum_{k=1}^{t} D^{-\ell_k}\right) \in g(\mathcal{D}^*)$ for all $t > 1$ and there exists an instantenous code.

Theorem

The expected length L of any instantaneous D-ary code for a random variable X is greater than or equal to the base D entropy, i.e.

$$L \geq H_D(\mathsf{X}).$$

The equality holds iff $D^{-\ell(x)} = p_{\mathsf{X}}(x)$ for all $x \in \mathcal{X}$.

Proof.

$$\begin{aligned}
L - H_D(\mathsf{X}) &= \mathbf{E}\left[\ell(\mathsf{X}) - \log_D \frac{1}{p_{\mathsf{X}}(\mathsf{X})}\right] \\
&= \mathbf{E}\left[\log_D \frac{p_{\mathsf{X}}(\mathsf{X})}{D^{-\ell(\mathsf{X})}}\right] \\
&= \mathbf{E}\left[\log_D \frac{p_{\mathsf{X}}(\mathsf{X})}{p_{\mathsf{Y}}(\mathsf{X})}\right] + \log_D \frac{1}{\sum_{\tilde{x}} D^{-\ell(\tilde{x})}} \quad \text{where } p_{\mathsf{Y}}(x) = \frac{D^{-\ell(x)}}{\sum_{\tilde{x}} D^{-\ell(\tilde{x})}} \\
&= D(p_{\mathsf{X}} \| p_{\mathsf{Y}}) + \log \frac{1}{\sum_{\tilde{x}} D^{-\ell(\tilde{x})}} \\
&\geq 0 \qquad \text{by ext. Kraft ineq. and nonnegtaivity of KLD}
\end{aligned}$$

Inequality in the last line is an equality iff $p_{\mathsf{X}}(x) = D^{-n}$ for some $n \in \mathbb{Z}_+$ for all $x \in \mathcal{X}$.

A p_{X} is D-adic iff $p_{\mathsf{X}}(x) = D^{-n}$ for some $n \in \mathbb{Z}_+$ for all $x \in \mathcal{X}$. $\square$

Shannon Code ($\ell(x) = \lceil \log_D \frac{1}{p_X(x)} \rceil$)

Theorem

For any discrete source X, *instantaneous source codes for D-ary alphabet satisfies*

$$H_D(\mathrm{X}) \leq L^\star \leq H_D(\mathrm{X}) + 1 \qquad \textit{where} \quad L^\star = \inf_{\mathcal{C}} L(\mathcal{C})$$

Proof.

Lower bound evident because $H_D(\mathrm{X}) \leq L(\mathcal{C})$ holds for all source codes $\mathcal{C}$.
To establish the upper bound we consider the Shannon Code:

- If $\ell(x) = \lceil \log_D \frac{1}{p_X(x)} \rceil$ then $\sum_x D^{-\ell(x)} \leq \sum_x D^{-\log_D \frac{1}{p_X(x)}} = \sum_x p_X(x) = 1$.
- Ext. Kraft's inequality is satisfied and $\exists$ an instantaneous code with these codeword lengths.
- $\ell(x) \leq \log_D \frac{1}{p_X(x)} + 1 \Rightarrow L \leq \mathbf{E}\left[\log_D \frac{1}{p_X(\mathrm{X})} + 1\right] = H_D(\mathrm{X}) + 1$.

□

Corollary

For any joint distribution on $\mathrm{X}_1, \ldots, \mathrm{X}_n$, *the minimum expected codeword length per source symbol* $L_n^\star$ *for D-ary alphabet satisfies*

$$\frac{H_D(\mathrm{X}_1,\ldots,\mathrm{X}_n)}{n} \leq L_n^\star \leq \frac{H_D(\mathrm{X}_1,\ldots,\mathrm{X}_n)}{n} + \frac{1}{n}.$$

Shannon Code With *Suboptimal (Wrong)* Distribution

Theorem

If $\ell(x) = \lceil \log_D \frac{1}{p_Y(x)} \rceil$ *then there exists a* D*-ary instantaneous source code* $\mathcal{C}$ *for* X *with these codeword lengths and*

$$H_D(\mathsf{X}) + D(p_{\mathsf{X}} \| p_Y) \leq L(\mathcal{C}) < H_D(\mathsf{X}) + D(p_{\mathsf{X}} \| p_Y) + 1$$

Proof.

$\exists$ an instantaneous code with these codeword lengths because Ext. Kraft's inequality is satisfied.

$$\log_D \frac{1}{p_Y(x)} \leq \ell(x) < \log_D \frac{1}{p_Y(x)} + 1 \qquad \text{by } \ell(x) = \lceil \log_D \frac{1}{p_Y(x)} \rceil$$

$$\mathbf{E}\left[\log_D \frac{1}{p_Y(\mathsf{X})}\right] \leq \quad L \quad < \mathbf{E}\left[\log_D \frac{1}{p_Y(\mathsf{X})}\right] + 1 \qquad \text{by } L(\mathcal{C}) = \mathbf{E}\left[\ell(\mathcal{C}(\mathsf{X}))\right]$$

On the other hand,

$$\mathbf{E}\left[\log_D \frac{1}{p_Y(\mathsf{X})}\right] = \mathbf{E}\left[\log_D \frac{p_{\mathsf{X}}(\mathsf{X})}{p_Y(\mathsf{X})} + \log_D \frac{1}{p_{\mathsf{X}}(\mathsf{X})}\right]$$
$$= D(p_{\mathsf{X}} \| p_Y) + H_D(\mathsf{X})$$

$\square$

$D(p_{\mathsf{X}} \| p_Y)$ is the excess redundancy due to a suboptimal distribution [Gal79, Rya79, DLG80].

Minimax Excess Redundancy and the Shannon Radius

Let W be a set of pmf's on $\mathcal{X}$. If $p_X \in W$ then what is the worst case excess redundancy?

$$S(W, p_Y) := \sup_{p_X \in W} D(p_X \| p_Y)$$ relative Shannon radius of W w.r.t. p_Y.

What is the best worst case excess redundancy?

$$S(W) := \inf_{p_Y} \sup_{p_X \in W} D(p_X \| p_Y)$$ Shannon radius of W.

Is the infimum always achieved? Yes.

Lemma ([Gal79, Rya79, DLG80])

If $S(W) < \infty$, then $\exists! p_{Y^*}$ satisfying $S(W) = S(W, p_{Y^*})$, i.e. $S(W) = \sup_{p_X \in W} D(p_X \| p_{Y^*})$, called the Shannon center of W.

Lemma will be proved later.

The Shannon center of a W need not to be a member of W in general.

If W is closed and convex, then the Shannon center of W is a member of W.

Theorem (McMillan Inequality)

Any uniquely decodable D-ary code for a finite source alphabet must satisfy the Kraft inequality

$$\sum_{x=1}^{m} D^{-\ell(x)} \leq 1.$$

Proof.

$$\begin{aligned}\left(\sum_{x=1}^{m} D^{-\ell(x)}\right)^k &= \sum_{x_1\in\mathcal{X}} \cdots \sum_{x_k\in\mathcal{X}} D^{-\ell(x_1)} \cdots D^{-\ell(x_k)} \\ &= \sum_{x_1^k\in\mathcal{X}_1^k} D^{-[\ell(x_1)+\cdots+\ell(x_k)]} \qquad (*)\end{aligned}$$

Let $a_k(l)$ be the # of length l codewords in the k fold extension the code.

$$\begin{aligned}\sum_{x_1^k\in\mathcal{X}_1^k} D^{-[\ell(x_1)+\cdots+\ell(x_k)]} &= \sum_{l=1}^{k\ell_{\max}} a_k(l) D^{-l} && \text{where } \ell_{\max} = \max\{\ell(1), \cdots, \ell(m)\} \\ &\leq \sum_{l=1}^{k\ell_{\max}} D^l D^{-l} && \text{the extension is non-singular thus } a_k(l) \leq D^l \\ &= k\ell_{\max} && (**)\end{aligned}$$

$$\sum_{x=1}^{m} D^{-\ell(x)} \leq (k\ell_{\max})^{1/k} \qquad \forall k \in \mathbb{Z}_+ \text{ by } (*) \text{ and } (**)$$

Thus $\sum_{x\in\mathcal{X}} D^{-\ell(x)} \leq 1$ because $\inf_{k\in\mathbb{Z}_+} (k\ell_{\max})^{1/k} = \lim_{k\uparrow\infty} (k\ell_{\max})^{1/k} = 1$. □

Theorem (Extended McMillan Inequality)

Any uniquely decodable D-ary code for a countable source alphabet must satisfy the Kraft inequality

$$\sum_{x \in \mathcal{X}} D^{-\ell(x)} \leq 1. \qquad (***)$$

Conversely given a $\ell : \mathcal{X} \to \mathbb{Z}_+$ satisfying this inequality, there exists an instantaneous, hence uniquely decodable code with these word lengths.

Proof.

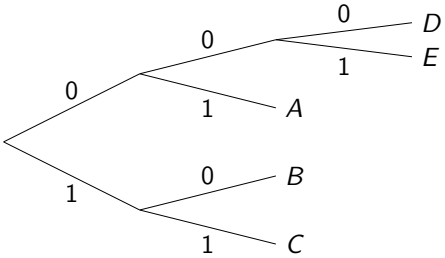
We prove the theorem by contradiction.

- Let us assume that $\sum_{x \in \mathcal{X}} D^{-\ell(x)} > 1$.
- Then there exists a finite set $\mathcal{U} \subset \mathcal{X}$ s.t. $\sum_{x \in \mathcal{U}} D^{-\ell(x)} > 1$.
- Then there exists a finite uniquely decodable code that violates Kraft inequality. This cannot be true because of McMillan inequality we have just proved.

If $\ell(x)$'s satisfy (***), there exists an instantenous code by extended Kraft inequality, which is uniquely decodable by definition. □

A Binary Huffman Code ([CT06, Example 5.6.1])

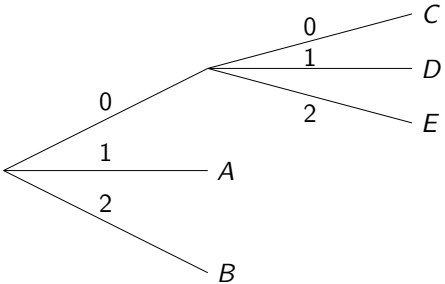
Codeword	X	Probability				
01	A	0.25	0.30	0.45	0.55	1
10	B	0.25	0.25	0.30	0.45	
11	C	0.20	0.25	0.25		
000	D	0.15	0.20			
001	E	0.15				



$$\mathbf{E}[\ell] = 0.15 * 3 + 0.15 * 3 + 0.2 * 2 + 0.25 * 2 + 0.25 * 2$$
$$= 2.3$$

A Ternary Huffman Code ([CT06, Example 5.6.2])

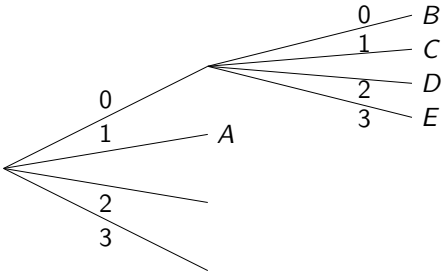
Codeword	X	Probability		
1	A	0.25	0.50	1
2	B	0.25	0.25	
00	C	0.20	0.25	
01	D	0.15		
02	E	0.15		



$$\mathbf{E}[\ell] = 0.15 * 2 + 0.15 * 2 + 0.2 * 2 + 0.25 * 1 + 0.25 * 1$$
$$= 1.5$$

A Non-Huffman Code ?

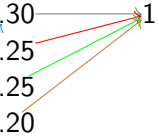
Codeword	X	Probability		
1	A	0.25	0.75	1
00	B	0.25	0.25	
01	C	0.20		
02	D	0.15		
03	E	0.15		

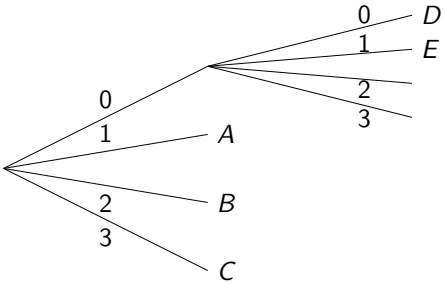


$$\mathbf{E}[\ell] = 0.15 * 2 + 0.15 * 2 + 0.2 * 2 + 0.25 * 2 + 0.25 * 1$$
$$= 1.75$$

A Huffman Code

Codeword	X	Probability
1	A	0.25
2	B	0.25
3	C	0.20
00	D	0.15
01	E	0.15
02	Dummy1	0.00
03	Dummy2	0.00





$$\mathbf{E}[\ell] = 0.15 * 2 + 0.15 * 2 + 0.2 * 1 + 0.25 * 1 + 0.25 * 1$$
$$= 1.30$$

A D-ary tree is full iff each node in the tree has either 0 or D children

Lemma

A D-ary tree is full iff it has $1 + k(D-1)$ leaves for some $k \in \mathbb{Z}_+$. Why?

In order to ensure that the empty leaves are as far away from the root as possible, i.e. the expected length is as small as possible, we add n auxiliary source symbols with zero probability to the M non-auxiliary source symbols to make the total number of source symbols $1 + k(D-1)$.

$$1 = M + n \mod D-1 \qquad\qquad \ell := M \mod D-1$$

$$n = \begin{cases} 1 & \text{iff } \ell = 0 \\ 0 & \text{iff } \ell = 1 \\ D-\ell & \text{iff } \ell \in \{2, \ldots, D-2\} \end{cases} \quad \text{and} \quad D - n = \begin{cases} D-1 & \text{iff } \ell = 0 \\ D & \text{iff } \ell = 1 \\ \ell & \text{iff } \ell \in \{2, \ldots, D-2\} \end{cases}$$

$$D - n = 2 + (M - 2 \mod D-1).$$

$D - n$ non-auxiliary source symbols are combined in the first step of Huffman procedure.

Remark

Huffman procedure can be used to minimize $\sum_x w(x)\ell(x)$ where $w(x)$ is the weight, rather than the probability, of the source symbol $x \in \mathcal{X}$.

Thomas M. Cover and Joy A. Thomas.
Elements of information theory.
Wiley-Interscience, New York, NY, 2 edition, 2006.

L. D. Davisson and A. Leon-Garcia.
A source matching approach to finding minimax codes.
IEEE Transactions on Information Theory, 26(2):166–174, Mar 1980.

R. G. Gallager.
Source coding with side information and universal coding.
Tech. Rep. LIDS-P-937, MIT Lab. Inform. Decision Syst, Cambridge, MA, September 1979.
`http://web.mit.edu/gallager/www/papers/paper5.pdf`.

B. Ryabko.
Encoding of a source with unknown but ordered probabilities.
Problems of Informution Transmission, 15(2):134–138, October 1979.